INNOTRACK INTERNSHIP PROGRAM - 2026

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- **Project Name – Air Quality Index Predictor**
- **Theme – Smart City**
- **VTU ID – 28149**

Air Quality Index Predictor – Smart City

INTRODUCTION:

Air pollution is one of the major environmental challenges affecting public health and urban sustainability worldwide. The Air Quality Index (AQI) depends on factors such as particulate matter (PM2.5 and PM10), carbon monoxide, sulfur dioxide, nitrogen dioxide, ozone levels, and weather conditions. Predicting AQI can help authorities monitor pollution levels and implement effective environmental management strategies. The Smart City Air Quality Index Predictor is a Machine Learning based web application that predicts AQI levels using historical air quality and environmental data. The system combines predictive analytics and a user-friendly web interface to provide real-time AQI predictions and support data-driven decision-making for healthier and more sustainable urban environments.

Idea :

- Develop a machine learning-based system to predict Air Quality Index levels.
- Support environmental monitoring and Smart City initiatives through data-driven insights.

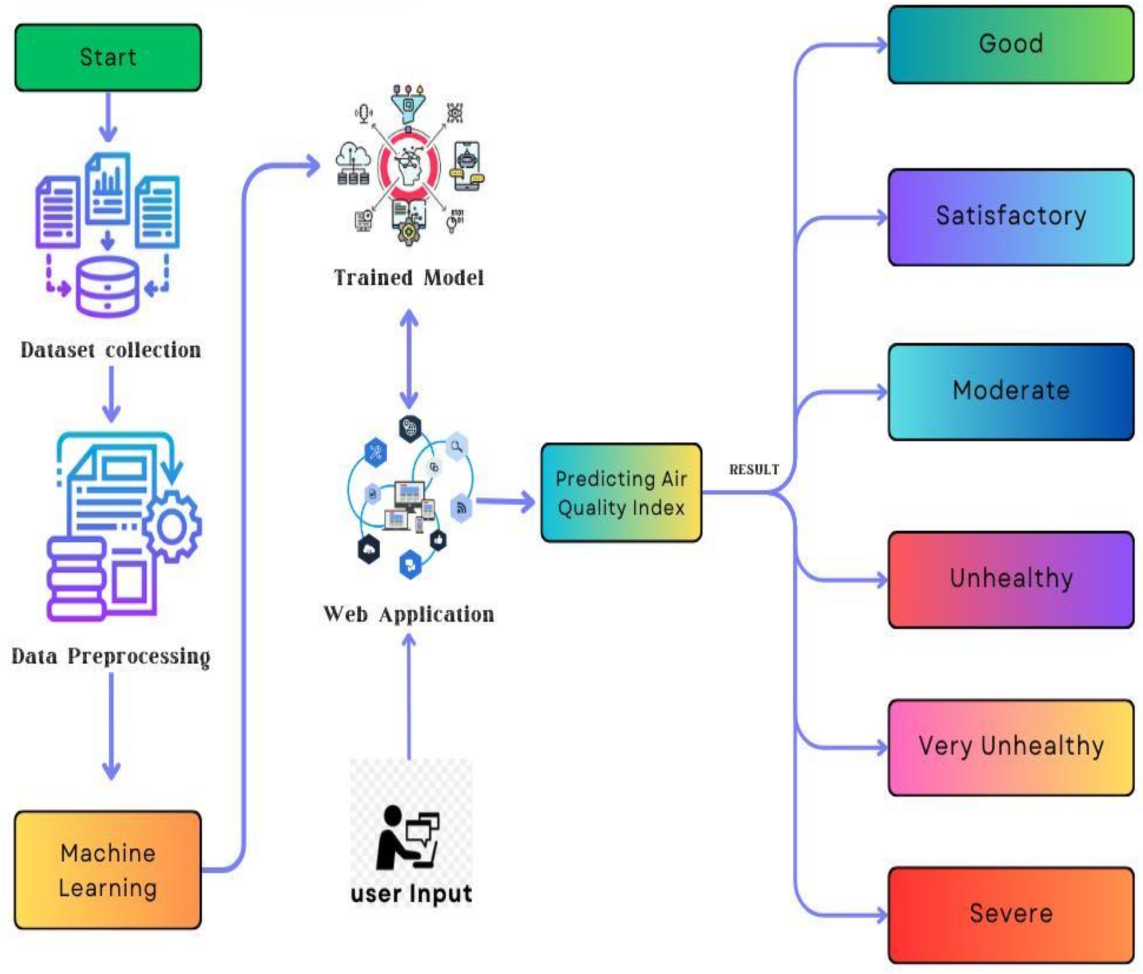
Solution :

- Built a Gradient Boosting model to predict AQI categories as Good, Moderate, Poor, Very Poor, or Severe
- Developed a Flask web application with real-time prediction, history tracking and analytics features.

Uniqueness:

- Real-time Air Quality Index prediction
- Interactive analytics dashboard
- Prediction history tracking and filtering
- User authentication and profile management
- Data-driven decision support for environmental authorities
- Centralized platform for air quality analysis and AQI assessment
- Secure storage of prediction records using SQLite
- Confidence score generation for each prediction

TECHNICAL APPROACH



Technologies to be Used:

- **Frontend** – HTML, CSS3, JavaScript
- **Backend** – Python, Flask Framework
- **Database** – SQLite
- **Tools** – Visual Studio Code , GitHub, Render
- **ML** - Gradient Boosting Regressor

Methodology:

- Processed air quality and environmental data and trained a Gradient Boosting Regressor model to predict AQI levels.
- Integrated the trained model into a Flask web application for real-time AQI prediction, history tracking and analytics.

FEASIBILITY AND VIABILITY

- Developed using open-source tools and technologies.
- Requires minimal hardware and software resources.
- Low development, deployment and maintenance cost.
- Easy to integrate with existing environment monitoring systems.
- Supports real-time Air Quality Index prediction.
- Suitable for Smart City and environmental management applications.
- Provides practical value for environmental authorities and analysts.
- Creates opportunities for future expansion into IoT and GIS-based applications.
- Reduces manual effort in air quality assessment and monitoring.

CHALLENGES FACED

- Handling missing and inconsistent values in the air quality dataset.
- Management variations in pollutant concentrations and environmental parameters.
- Integrating the trained machine learning model with the Flask web application.
- Designing a professional and responsive user interface.
- Implementing user authentication and profile management features.
- Ensuring accurate real-time AQI predictions and database storage
- Creating meaningful analytics and visualization dashboards.
- Maintaining consistency between training, testing, and deployment environments.

FUTURE ENHANCEMENTS

- Integrating real-time air quality data from IoT sensors and monitoring stations.
- Enhancing AQI prediction accuracy using advanced machine learning and deep learning models.
- Integrating weather forecasting data for improved pollution prediction.
- Developing a mobile application for AQI monitoring and alerts.
- Implementing location-based air quality tracking and notifications.
- Adding GIS-based pollution mapping and hotspot detection features.
- Creating advanced analytics and visualization dashboards.
- Deploying the system on cloud platforms for large-scale environmental monitoring.

IMPACT AND BENEFITS

- Enables early assessment of air quality levels using machine learning.
- Supports environmental authorities in making informed decisions.
- Provides instant AQI predictions for effective pollution monitoring.
- Helps identify pollution pattern and environmental trends from historical data.
- Improves environmental planning and resource allocation.
- Supports Smart City development through data-driven solutions.
- Provides an easy-to-use platform for air quality analysis and reporting.
- Demonstrates the practical application of AI and Machine Learning in environmental management.
- Enhances public awareness of factors affecting air quality and pollution.

CONCLUSION

The Smart City Air Quality Index Predictor successfully demonstrates the integration of Machine Learning and Web Development to address a real-world environmental challenge. The system effectively predicts Air Quality Index levels based on key environmental parameters such as PM2.5, PM10, carbon monoxide, nitrogen dioxide, sulfur dioxide, ozone concentrations, and weather conditions, providing quick and reliable insights for better decision-making. This project highlights the importance of data-driven solutions in environmental monitoring and supports the vision of Smart Cities. By combining predictive analytics, a user-friendly interface, and database integration, the system offers a complete end-to-end solution that can be further enhanced for real-time monitoring and large-scale environmental management in the future.

THANK YOU